# Ideas to Steal

A literature review for Stitchwort, summarising the seven SW-894 research registers

Stitchwort project, board ticket SW-894

30 October 2026

**Abstract**

Stitchwort is a small, total, dependently typed functional language written in Rust, with algebraic effects and a small trusted kernel. Between 29 and 30 October 2026 it was the subject of a literature review in seven clusters: effects and handlers, elaboration, proof automation, the kernel and type theory, inductive types and recursion, infrastructure and trust, and compilation and runtime. The window was 2022 onwards. Across the seven registers 449 sources were screened and 160 carded; 154 of the cards rest on a full text read for the review and 6 were not read. A first attempt on 29 October wrote several registers from abstracts. Every register was then redone or audited against full texts on 30 October, and this summary cites only sources read in full.

The review has already changed Stitchwort. Porting a 2026 kernel-bug shape from Lean4Lean found a real soundness bug, an unchecked universe-level overflow, which was fixed (SW-902). Fourteen further must-reject programs, shaped like 2026 bugs in other provers, are all rejected. The confluence argument for the extra `Nat.add` rules now rests on the Taming of the Rew. Two kernel hazards are recorded as decisions against future work (SW-93, SW-55). The rest of the findings point the same way. The literature broadly confirms Stitchwort's architecture: an untrusted elaborator, certificates checked by the kernel, features fixed when a declaration is admitted. What it adds is mostly sharper tests and oracles, a handful of kernel restrictions, proof automation that stays inside the elaborator, and one elaborator redesign (a single postponement queue). Section 12 gives a prioritised list and Section 13 the decisions still open.

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# 1 Stitchwort in brief

This section is for a reader without access to the repository. Everything here is from the project's own documents (`README.md`, `DESIGN.md`, `docs/core.md`) as they stand on `main` on 30 October 2026, and from the registers' descriptions of the code.

**The language.** Stitchwort is a strict, pure, dependently typed functional language, implemented in Rust (the `crates/` directory: `syntax`, `elab`, `core`, `rt`, `js`, `native`, `driver`, `cli` and others). It combines four ideas that usually live in separate languages: functions terminate by default, and anything that might not says so in its type; dependent types let programs carry proofs about themselves; effects are tracked in types and implemented by handlers, in direct style; and definitions are identified by a hash of their contents, so tests and proofs whose dependencies haven't changed are never re-run. `Prop` is impredicative and erased at run time; universes are numbered and not cumulative; the kernel has axiom K. The one axiom is `Quot.sound`, and `stitchwort axioms` reports whether a definition depends on it.

**Abilities.** Effects are called *abilities*, after Unison. A computation with abilities is translated into data, `Effect.Comp`, with constructors `Done` and `Call`; a handler is a fold over it (`Effect.foldComp`), and sequencing is `Effect.bind` (`crates/elab/src/std/core/Effect.sw:12-28`). Handlers are deep and resumptions may be used any number of times (decision D1162). Because handlers are ordinary folds, the kernel's own positivity and termination checks prove that handled programs terminate; what remains trusted is that the translation is faithful to the source semantics.

**The kernel.** A small trusted kernel (`crates/core`) re-checks every declaration the elaborator produces. It has case trees with size-change termination rather than recursors, an untyped conversion checker with eta for functions, two extra conversion rules for `Nat.add` (R3/R4, pinned by content hash), and a short list of natives (fast implementations of standard definitions). The elaborator's unifier is untyped and may be incomplete; that is safe because the kernel re-checks the result.

**The elaborator.** The elaborator (`crates/elab`) does pattern unification with pruning, keeps glued calls (definitions compared by arguments before unfolding), infers implicit arguments, solves instance obligations in a per-declaration loop without backtracking, and has records with literals, projections and a field-name rule for literals whose type is unknown. It postpones lambda and method-section arguments within one application, and since SW-896 puts aside unification pairs stuck on an unknown.

**Proof tools.** Proofs are terms, written with `match`, `J`, `rewrite`, and a few automation words: `rfl`; `cases` (case splits with closers: computation, constructor clash, injectivity, a hypothesis, `omega`); `omega` and `ring`, which search outside the kernel and produce a certificate the kernel checks; `auto`, which uses `Decidable` instances and hypotheses in scope; and `decreasing` for recursion on a measure, with `Nat` fuel rather than `Acc`. A counterexample search runs on a failing theorem, filters random inputs by its hypotheses, and is advisory only. A `?` hole leaves a goal and checking continues.

**Compilation.** Programs run on a fast runtime (`crates/rt`): an erased IR, register bytecode with moves on last use and Perceus/Lean-style reuse tokens, small numbers inline with a bignum fallback, tail recursion modulo context by destination passing, and state handlers run as loops. Native code is generated with Cranelift (`crates/native`) and keeps the machine's reference counting decisions. A JavaScript backend serves `run --js` and the extraction of proven code as an ES

module with TypeScript types. A reference evaluator (`run --reference`) runs programs on the type checker's own evaluator. A browser playground runs the machine as WebAssembly.

**The corpus.** A standard library (`std`); examples, including must-reject examples in `examples/soundness/` (107 of them after 30 October); Advent of Code 2024 and 2025 with proofs (`aoc/`); and a roguelike, `games/rogue`, with its own proof notes. Tests run in tiers (lane, merge, daily, full), with fuzzers for the parser, formatter, elaborator and kernel, and differential tests across the four ways of running a program.

## 2 Method

### 2.1 The seven clusters

The plan, in SW-894, divided the field into seven clusters, each with board hooks:

1. **Kernel and type theory**: rewrite rules, definitional proof irrelevance, record eta, observational type theory, quotients, universe polymorphism.
2. **Inductive types and recursion**: nested and inductive-inductive types, size-change termination, well-founded recursion.
3. **Elaboration**: unification, postponement, implicit arguments, glued evaluation, error messages.
4. **Proof automation**: rewriting, decision procedures, e-graphs, counterexample search.
5. **Effects and handlers**: rows, evidence and capability passing, handler termination, effects with dependent types.
6. **Compilation and runtime**: erasure, reference counting and reuse, interpreters, JS and WebAssembly.
7. **Infrastructure and trust**: content-addressed code, incremental compilation, testing and fuzzing of checkers, agent-driven development.

Each cluster has one register in `docs/research/`, with the same parts: method and queries, screened list, cards (about five lines per kept source), deep reads of the top three to five, ranked "steal next", proposed tickets, and at most a few decisions for Matt.

### 2.2 Window

Matt set the window at 2022 onwards (D1831), with snowballing inside it. Older work appears only where in-window work or Stitchwort's own code relies on it, and is marked *classic* in the registers and in this summary (for example the Taming of the Rew, 2021, and Counting Immutable Beans, 2019).

### 2.3 The reading protocol

After the failure described next, every register follows one protocol:

- **Full text or nothing.** PDFs were read with `pdftotext` from `nix shell nixpkgs#poppler-utils`; source files, manual pages and issues were read whole. For a long theory paper, "read" means the prose, definitions, examples, evaluation and conclusions, with proofs skimmed; each card names the sections read.
- **Counts first.** Every register opens with screened, carded, read in full and not read.
- **NOT READ supports nothing.** Every card says "Full text: read (source)" or "NOT READ". A card not read cannot be deep-read, ranked or used for a recommendation.

- **The paywall rule.** Sources were tried in order (arXiv, authors' pages, DBLP, Semantic Scholar, Dagstuhl, HAL, ACM, the Internet Archive). A full text that couldn't be reached goes in a "for Matt to fetch" table only if it would change something; the rest are listed as not reached.
- **No personal data** was sent to any service.

### 2.4 The 29 October failure and the 30 October redo

The first attempt, on 29 October, went badly; SW-897 records how. No PDF tool was installed and the web-fetch tool could not read PDFs, and nobody checked before the lanes started, so the elaboration, proof-automation and kernel registers were written largely from abstracts. Each lane listed its unread sources at the end of its report, and the orchestrator relayed the findings and dropped the caveat. Up to four research lanes ran at once and exhausted the session's shared web-search pool, and one lane's claim that the budget had run out was relayed without checking. The orchestrator once sent Matt's email address to an open-access API. The kernel redo's readers later found that the first kernel register misstated four papers.

Matt paused the work (SW-897) and restarted it on 30 October one lane at a time (D1974), with the protocol above, a tested `pdftotext` toolchain, reserved decision numbers and a planned search budget. The elaboration, proof-automation and kernel registers were redone from full texts; the effects, inductives, infrastructure and compilation registers were audited card by card against full texts. Each redo also lists the 29 October claims it found wrong: five in effects, six each in elaboration, proof automation and kernel. The audits found one wrong card each in effects (a steal), infrastructure (numbers taken from an anonymous submission copy) and compilation (a thesis example attributed to Lean's library), and none in inductives, though one of its assumptions about Stitchwort was wrong.

### 2.5 Counts

Table 1 gives each register's counts as its first line states them. Some sources are carded in two registers (for example Lean's functional induction, the Hazel error-recovery paper and the PLDI 2022 inductive-relations paper), so the totals count cards, not distinct works.

Web searches were few in the redo: five each for elaboration and kernel, six for proof automation, one for the effects audit, none for the inductives audit and twelve for the infrastructure audit, which finished the searching its 29 October version had been cut short of.

## 3 What has already been acted on

Table 2 lists what the review has already changed, with the commits and tickets that did it. Each item was checked in `git log` and on the board on 30 October.

Table 1: Counts per cluster, from each register's first line (30 October 2026).

| Cluster | Screened | Carded | Read | Not read | How checked |
|---|---|---|---|---|---|
| Effects and handlers | 75 | 11 | 11 | 0 | Audited: 6 OK, 4 minor, 1 wrong (fixed). The 11 read are 10 papers and Lean's manual (C3). |
| Elaboration | 64 | 20 | 20 | 0 | Redone from full texts; 6 old claims wrong. |
| Proof automation | 72 | 29 | 25 | 4 | Redone from full texts; 6 old claims wrong. |
| Kernel and type theory | 62 | 27 | 26 | 1 | Redone from full texts; 6 old claims wrong. |
| Inductive types and recursion | 56 | 21 | 21 | 0 | Audited (19 cards): 11 OK, 8 minor, 0 wrong; 2 cards first NOT READ were read. |
| Infrastructure and trust | 68 | 29 | 29 | 0 | Audited (24 cards): 18 OK, 5 minor, 1 wrong; searching finished; C3 (Eris) read later from the PDF Matt fetched. |
| Compilation and runtime | 52 | 23 | 22 | 1 | Audited (21 cards): 12 OK, 8 minor, 1 wrong; C12 read at the audit. |
| **Total** | **449** | **160** | **154** | **6** | |

Table 2: Already acted on.

| Item | What was done | References |
|---|---|---|
| Universe-level overflow (soundness bug) | Found by the lane porting 2026 kernel-bug shapes (kernel register T1), on Lean4Lean's `looseBVarRange` overflow shape [22]. `def wrap: Type 0 := Type 4294967295` was accepted: levels were `u32` and the step to the next universe wrapped. Fixed: `MAX_LEVEL` is 65535 and `Sort::succ` returns no successor for it (`crates/core/src/term.rs:33`, `:40`); parser, elaborator and cache limits; five `universe-*.sw` examples and `crates/core/tests/universe_levels.rs`. | SW-902 (D1984); e1d444d7, da80af16, d8d36404, d3c2dee4; merged 65f6a236. |
| 14 must-reject tests for 2026 kernel-bug shapes | Programs for Rocq CVE-2026-72703 and 72705, Arena `k-rec-conv`, `eta-unit-like-proof-field` and `forged-quot-ind`, and Lean's order-dependent cache (lean4#14806); all rejected (`E-TERMINATION`, `E-REFL`, `E-DUPLICATE`), about 0.13 s of CPU each. A kernel-level test builds the same terms by hand (`crates/core/tests/kernel_bugs_2026.rs`, 6/6). | Kernel register T1 (D1983); 3208ab72, 1ae69c78; merged f38544f6. |

| Item | What was done | References |
|---|---|---|
| `+` confluence on the Taming of the Rew | `docs/proposals/nat-add-recursion.md` §9.4 now argues confluence of R1–R4 by the triangle property [28] instead of Newman's lemma; a new assumption A1′ (the core's own triangle property); test `the_parallel_rule_is_reached` (`crates/driver/tests/nat_add_rule.rs:160`) checks that the evaluator reaches the parallel rule's right side. | Kernel register T4 (D1983); c1084edd; merged f9521afe. |
| Recursive propositions (SW-93 warning) | D1980 (Matt): refuse eliminating a self-referential proposition into data, as part of SW-93, after measuring that nothing in the corpus does it. The warning is in SW-93's body. | D1980; ba9ba8a7. |
| Record eta reminders | D1981 (Matt): once proofs are irrelevant, record eta only for records with a field that isn't a proof. Reminders placed on SW-93, SW-55, SW-789 and SW-568. | D1981; ba9ba8a7. |
| Counterexample search misses | SW-903 records that `check --counterexamples` misses a false theorem silently when its hypothesis holds only for small inputs; re-run by the orchestrator on `main`. | SW-903 (backlog); 9094029f. |
| Congruence closure in `cases` | SW-904 records the idea. D2000 (Matt): not prioritised yet; first survey existing proofs for steps it would remove. | SW-904 (backlog), D2000; 2a66a5f9. |
| One postponement queue | D1990: a design note, `docs/proposals/postponement.md`, comparing Lean 4, Agda, Idris 2, Rocq and OmniML. D2012 (Matt): design for the best end state, so draft 2 makes one queue in the single elaboration pass, with prototype measurements and a brief for an external reviewer. The review (approve with revisions) is worked into draft 3 (D2015), merged. | SW-817 (doing); 7b751873 / 8cd3cdd6; c35e458b / bb9e432a; 3d0036f8. |
| SW-805, field-name rule | D1991: keep the rule as a name-resolution rule; it was merged as a last resort, used only where nothing else gives the literal's type. | SW-805 (done); 8f52025b; merged afea8395. |

| Item | What was done | References |
|---|---|---|
| `decide.md` (D1971–D1973) | Design note for SW-800 and SW-807: extend `cases` with a decision closer and a split of a finite variable, derive a `Finite` interface, never `native_decide`. Merged; the proof-automation register finds that D1971 stands, and the inductives audit withdrew its D1912 in favour of D1972. Its review page is published; the decisions await the external review. | 88e9104b, 0ddc8e2c; SW-800 (next). |
| Handler termination in `DESIGN.md` | Effects register T1 (D1852): DESIGN.md described handler termination as a tracked conjecture; it now says the kernel checks it and names the two trusted items. | SW-899; 4e33920c. |

# 4 Effects and handlers

**Register:** `docs/research/effects.md`. 75 screened, 11 carded, all read; audited.

## 4.1 Key findings

- **Monad laws by conversion have a price.** Yang and Wu get the monad laws judgementally for handlers carried by lawless raw monads, by giving up the law that handling commutes with `let`, since with lawless monads it "would result in inconsistency" (Remark 2.11) [142]. Their calculus is System F$\omega$ with effects, not dependently typed, and its §6 leaves open whether a dependently typed language can have all the laws judgementally. Their model is a CPS encoding in which the laws hold by $\beta$.
- **Hoare triples that yield equations.** Kidney, Yang and Wu's H-elim turns a triple into a program equation that can be substituted in any context [57]. Its hard part comes from quotients; on Stitchwort's unquotiented `Comp` it is an easy induction, but it still needs `Quot.funext` at `Call` and, the audit found, it helps neither rogue's "gap 2" nor SW-548.
- **Specifications as predicate transformers.** Lean's `Std.Do`, `mvcgen` and `vcgen` give each program a weakest-precondition reading and reduce triples to verification conditions with no weakest preconditions left [79]. `vcgen` accepts any program type with a `WP` instance, including syntax trees.
- **Declared fast paths.** Lexa declares handlers tail-resumptive or abortive and runs them in place, "reducing running-time complexity from quadratic to linear in some cases" [96]; it treats multi-shot resumptions as a non-goal. A per-effect strategy chosen by invocation rate gives exceptional effects zero overhead [95].
- **TRMC through effectful callbacks.** Koka keeps an effectful `map` in constant stack either by copying a shared context or by running a second version whenever a multi-shot handler is in scope [84].
- **Termination.** Hamana proves a handler strongly normalising by the argument the kernel uses, that `M[x]` is smaller than `get(x.M[x])` [47].
- Smaller points: messages should never show a hidden effect variable [128]; Koka closes rows at `let`, a related but different rule from Stitchwort's D1124 [52]; interaction-tree specifications are a further-off fit, for coinductive programs [121]; Lean's `do` translation targets transformers, later replaced by CPS versions for speed [133].

### 4.2 What Stitchwort could take

The register's ranking, by value against effort (it was not re-ranked at the audit):

1. ~~Retire the handler-termination conjecture text~~ *(done, SW-899).*
2. `Effect.bindCongAlways`, now narrowed to a small std lemma for rewriting a continuation under a fact the computation guarantees.
3. A handler fast-path assertion or report (T5): `crates/rt/src/fusion.rs`'s recognisers return `Option` with no reason (`fusion.rs:242`, `:269`, `:482`), so SW-861's slowdown (9.4 s against 2.2 s) passed silently. Re-measure SW-861 first: a later commit may already fuse its natural form [INFERRED].
4. Handler specifications as predicate transformers (T4, D1851).
5. The monad laws of `Effect.bind` and `Effect.foldComp` as trusted conversion rules (T3, D1850): a kernel change, design note only. Unlike R3/R4 these rules are higher-order and need funext, so the `Nat.add` confluence argument does not carry over; the note must weigh them against a CPS representation, which would lose induction on computations.
6. Messages never show a bare row unknown.
7. Guarded TRMC for an effectful `map` (SW-868 option (b')).
8. A per-ability runtime strategy (research).
9. Modal effects instead of `{e}` on higher-order functions (research; conflicts with "a plain arrow means pure and total").

### 4.3 Acted on, and open

SW-899 rewrote DESIGN.md's handler-termination text (D1852, closed). Open: D1850 and D1851 (Section 13).

## 5 Elaboration

**Register:** `docs/research/elaboration.md`. 64 screened, 20 carded, all read; redone.

### 5.1 Key findings

- **One postponement queue.** Lean postpones any elaborator that needs an unknown expected type (record instances, dot notation on an unknown type) and resumes them in a fixed ladder: resume while progress is made, then without postponing, then default instances, then report; it also has a body-first `let` [68]. OmniML specifies the idea: a construct waits on the shape of one type, each unknown keeps a wait list, and a wait never discharged is an error [106]. ExEl runs every solver over one constraint store after each declaration [91]. Stitchwort postponed only lambda and method-section arguments within one application (`crates/elab/src/check.rs:1903`, `apply_args_postponing`), so record literals (SW-817, SW-885) and let-bound lambdas (SW-847) failed at once.
- **Stitchwort's unifier is lossy for stuck calls.** Two forced calls of one definition with the same head have their spines unified as ordinary pairs, which may solve unknowns (`crates/elab/src/unify.rs:929-944`). That is the approximate rule: Coq and Lean use it by default, Agda only on request because it "will cause Agda to ignore some possible solutions" [4], and smalltt forbids it [62]. Polarity states the precise rule: congruence on arguments is safe for a neutral head, not a blocked one [90]. The kernel's re-check keeps it sound; nothing in `docs/core.md` says the elaborator may pick a different solution than the user meant.
- **SW-805's field-name rule is the one GHC kept.** GHC removed every type-directed field

rule but kept "only one datatype has all the fields", because it is decided during name resolution [46]; OmniML types a literal by the same closed-world reasoning [106].

- **Errors.** In the one controlled study, correct hints raised success and speed for Agda newcomers and incorrect ones did no measurable harm (56 participants, 412 submissions) [56]; flow-based messages showed no quantitative gain, and bare `?a` unknowns confused readers (117 participants kept) [13]; total recovery needs a type for every erroneous term [143].
- **Benchmarks.** smalltt's `idTest` (40 nested `id`s) and `pairTest` (30 nested `let` pairs) take 0.000 s in smalltt and time out in Agda, Coq, Lean and Idris; `vecTest` (960 elements) takes 0.078 s against 0.244 s in Lean (2021 versions) [62].
- Background: Lean's height-ordered unfolding applies only when neither side has metavariables [73]; `@[elab_as_elim]` infers a motive from the expected type [68]; pairs of values preserve unfoldings [60]; Rocq ignores mismatched types after postponing and relies on the final re-check [61]; expression-level `unfold` complements opaque definitions [44].

### 5.2 What Stitchwort could take

1. One declaration-level postponement queue with Lean's ladder (T1, D1990): now in design as SW-817.
2. Document and test the lossy rule for stuck calls (T2): small, no behaviour change.
3. SW-805 justified as a name-resolution rule, with an ambiguity message naming both types (T3).
4. smalltt's three benchmarks as elaboration probes (T4).
5. Motive inference for `T.induct` applications (T5).
6. Messages say which expression solved an unknown; never a bare `?n` (T6).
7. Keep both named and forced expected types in `check_inner` (SW-858).
8. Expression-level `unfold f in e` (SW-796, SW-798).
9. Lean's scope-argument constant approximation (measure first, SW-815).
10. Pattern-matching unification [112]: research, not now.

Not recommended: functions-as-constructors unification [89], complete search [105], contextual typing [141], an extensible-constraints architecture, and tabled resolution [118], which only matters with overlapping instances.

### 5.3 Acted on, and open

D1990 led to `docs/proposals/postponement.md`. Its first draft measured that the queueable `E-CANNOT-INFER` sites fire 1,073 times in accepted code, always caught, and proposed a second pass; D2012 made the end state one queue in the single pass. The draft 2 prototype changed one acceptance and 26 hashes (statements identical), and no error message. The external review (approve with revisions) is worked into draft 3 (D2015): queue invariants, a rule for kept receivers, a termination argument over a finite attempt tree, and tri-state unification as the expected follow-up. SW-805 is merged as a last resort (D1991). Open: the note's D2013, D2014 and D2016 (Section 13).

## 6 Proof automation

**Register:** `docs/research/proof-automation.md`. 72 screened, 29 carded, 25 read, 4 NOT READ; redone.

## 6.1 Key findings

- **Equalities among hypotheses.** Three probes fail with `cases` today [RUN]: two hypotheses equating a list with `Cons` and with `Nil`; `f a = c` from `a = b` and `f b = c`; and a local `(y: Nat) -> lookup = Some y -> y < 5` applied to `lookup = Some y0`. Congruence closure with a proof forest settles the first two [70, 117]; instantiating local ∀ facts settles the third. Selsam and de Moura's subsingleton rule handles proof arguments, and Stitchwort already has the lemmas it needs (axiom K, `Bool.uniqueEq`, `DecEq.uniqueEq`).
- **Lean's `grind`.** All proofs are by contradiction on a shared whiteboard; it produces ordinary proof terms; constraint propagation closes goals with no case split; case analysis is bounded and "not exhaustive"; it is "not designed for goals whose search space explodes combinatorially" [78, 102]. Statically predicting the right split is no better than random: on rescuable split failures stock `grind` gets 0%, learned models 30–53%, uniform random 57% and one-step lookahead 100% [135].
- **The counterexample search misses some false theorems silently.** `shortIncreasing (xs: List Nat) (h: increasing xs = True): xs.length < 6` is false, yet `check --counterexamples` prints "none in 10000 cases" [RUN]. The verdict is "gave up" only when hypotheses hold in under 10% of inputs (`crates/driver/src/counterexample.rs:82`); here they hold often, but only for short lists. Recent work builds generators from the hypothesis itself: Palamedes synthesises them from Lean `Bool` predicates (32 benchmarks; 80% synthesised in under 3 s) [42]; QuickChick orders hypotheses by "density" [99]; derived checkers are fuel-bounded semi-decision procedures, not `Decidable` instances [108]. Structural shrinking, the kind Stitchwort has, is competitive [55]; Plausible filters as Stitchwort does [66].
- **Case theorems from the definition.** Lean's `f.fun_cases` and `f.induct` add an equation for each discriminant, "just as if the user wrote `match h : x with ...`", and are generated on first use [69].
- **Show what search found.** Sledgehammer writes a one-line proof with instantiated facts in 70.0% of cases [11]; querysmt writes a self-contained script so the proof no longer needs the solver [26]; APOLLO's loop tries built-in tactics at each `sorry` [107].
- **External solvers are expensive.** lean-smt supports about 200 of cvc5's 662 proof rules and verifies 15,271 of 21,595 proofs (71%) [100]; Lean-auto reconstructs only through Duper [27, 111]; crush closes 693 of 754 conditions (91.9%) with reconstruction against `grind`'s 641 (85.0%) [48]; premise selection adds 21% more goals [145].
- Background: small proofs from congruence closure [37]; Lean's `omega` [71] and termination guessing [72]; reflective rewriting scales where proof-producing rewriting is quadratic in binders [45]; e-graph rewriting under binders is hard enough that lean-egg's authors moved to `grind` [114]; keyed matching for reflective `ring` [116].

## 6.2 What Stitchwort could take

1. A congruence-closure closer in `cases` (T1, D2000).
2. Report "hypotheses held only up to size $k$" in the counterexample search, and measure it over std and aoc (T2; SW-903).
3. Local ∀ hypotheses instantiated once against the classes (stage 2 of 1).
4. Generated `f.cases` (then `f.induct`) on demand (T4, D2001).
5. Lookahead over other atoms before `E-CASES-BUDGET` (T6).
6. Opt-in reports of what `cases` and `omega` used; "try automation at each `?`" (T5).
7. Hypothesis-directed inputs, only if 2 finds cases (T3).
8. Propagation over `Bool` connectives before splitting.
9. Suggest a `decreasing` measure on a termination failure.

10. Reflective rewriting, if `simp [rules]` is ever built.

Not recommended: SMT or ATP back ends, e-graph rewriting under binders, and bit-blasting (its card is NOT READ, and no program needs bit vectors).

### 6.3 Acted on, and open

SW-903 records the silent miss; SW-904 records congruence closure, with Matt's D2000: evidence from existing proofs first. On `decide.md`: `grind`'s own splitting is bounded too, which supports D1971; the literature adds propagation before splitting and lookahead as options. Open: D2001.

## 7 Kernel and type theory

**Register:** `docs/research/kernel.md`. 62 screened, 27 carded, 26 read, 1 NOT READ; redone.

### 7.1 Key findings

- **Recursive propositions and proof irrelevance.** If a proposition can be eliminated into data and holds a proof of itself in a field, definitional irrelevance lets any proof be unfolded once more, and conversion becomes undecidable [35, 41]. Lean is removing such large eliminations [2, 19]. Stitchwort's `E-PROP-ELIM` (`crates/core/src/check.rs:915-929`) does not ask whether a field mentions the type itself, so `proof-irrelevance.md`'s `Loop` is accepted. Rocq implements irrelevance in an untyped conversion with relevance marks, the same idea as SW-93's D1′ masks [41]. A cast that reduces on convertible endpoints breaks normalisation with an impredicative `Prop` [1]; Stitchwort's `J` computes only on `rfl`, so this is outside the kernel.
- **Record eta needs a field that isn't a proof.** Once proofs are irrelevant, a record of proofs is unit-like and eta on it breaks transitivity; Rocq requires a relevant field [85, 113], and Lean's kernel hit the same case [74].
- **R3/R4 have a published confluence argument that needs no termination.** The Taming of the Rew uses parallel plus, Stitchwort's four rules, as its example: with a fifth "parallel" rule they satisfy the triangle property, which gives confluence without normalisation [28]. The Rewster shows the naive typing check unsound under cumulativity, which Stitchwort lacks [86].
- **What broke other kernels in 2026.** Rocq's guard checker (CVE-2026-72703, 72705) [20]; Lean's order-dependent cache [103] and name-trusted native `Nat.add` [76]; the Arena's forged `Quot.ind` [74]. An agent team found seven proofs of `False` and three other bugs in Rocq, none in the official Lean kernel (four other bugs), two in nanoda and one in lean4lean [126]. Stitchwort's natives are bound by name and checked by type and arity only (`crates/core/src/native.rs:379-455`), the configuration Lean4Lean calls "technically unsound" [22]. Lean4Lean also reports that soundness bugs occur where "the developer did not even consider the error condition".
- **Quotients and erasure.** Danielsson proves that with quotients and erased transport, closed programs' extractions compute the right numeral even where source reduction is stuck (Theorem 11.3) [31]: the nearest published result to `map-equality.md` §7.4.5's "intended invariant", under different assumptions.
- **Mechanised checkers and their cost.** Lean4Lean runs 1.21–1.40 times slower than Lean's kernel [22]; MetaCoq's development is over 300 kLoC and found one kernel incompleteness, not a soundness bug [124]; Martin-Löf à la Coq is about 30 kLoC [3].
- Not recommended now, each sound research no ticket needs: observational equality [109,

110], universe polymorphism and cumulativity [12, 40, 125], a concurrent convertibility checker [30], definitional inversion without normalisation [23], typed confluence [36].

### 7.2 What Stitchwort could take

1. No large elimination from recursive propositions, measured first (T2, D1980).
2. Must-reject tests for the 2026 shapes (T1) *(done, 14 programs)*.
3. Record eta needs a non-proof field (T3, D1981).
4. §9.4 on the triangle property (T4) *(done)*.
5. Natives: check the defining equations at binding, or pin by hash (T5; SW-898).
6. One sentence in `docs/core.md`: memo tables are keyed by exact pairs, never closed transitively (T6). Stitchwort's memos already are: `CallMemo` (`crates/core/src/value.rs:2334`) and the equal-pairs record of `same_stuck_trees` (`value.rs:2736`).
7. Cite Danielsson's Theorem 11.3 in `map-equality.md` (T6).
8. Courant and Leroy's conversion probes (T7).
9. Inputs to a future universe-polymorphism note (research, not now).
10. Observational equality for funext and recursion through quotients (research, not now).

### 7.3 Acted on, and open

T1 and T4 are done; T1 found SW-902. D1980 and D1981 are decided and recorded on the tickets they constrain. SW-898 (pin the natives) is in the backlog. No kernel decision is open.

## 8 Inductive types and recursion

**Register:** `docs/research/inductives.md`. 56 screened, 21 carded, all read; audited.

### 8.1 Key findings

- **Stitchwort's big choices are confirmed.** Lean removed reducible well-founded recursion in 4.25 (#10714) [75], and Leroy's pearl supports recursion on a measure with proofs by induction on the argument [87]. Sized types remain impractical: backward-compatible size inference slowed one file of Coq's standard library, `MSetList`, from 15.1 s to 83.7 s (5.5×), "intrinsic and unavoidable" [25]; Agda's sized types are inconsistent because Agda proves $\infty < \infty$ [64].
- **Generated induction and case principles.** Lean's `f.induct` and `f.fun_cases` follow a function's branches, with one hypothesis per recursive call [69]. Breitner's warning: facts that exist only inside a termination proof aren't available to the induction principle ("The cake is a lie") [18]. Generated definitions checked afterwards are Stitchwort's route already; Ko et al. show generation costs far more than the definitions themselves (about 66 s against 5 s) [34, 59].
- **Automatic `attach`** rewrites `xs.map f` to its attached form so a recursive call inside `map` gets its membership fact [67].
- **Termination-measure suggestions.** GuessLex tries candidate measures and prints the one that works, or a table per call [72].
- **Codata.** Checking termination and productivity separately "is not enough" once inductive and coinductive types nest [50]; Lean added only coinductive predicates [77].
- Smaller points: Lean's `partial_fixpoint` reasons about possibly non-terminating functions [77]; derived enumerations in linear size [97]; derived checkers for relations [108]; represen-

tation of `Nat`-like families [51, 130]; constructors resolved by expected type [5, 81]; motive inference by pattern-matching unification [112].

### 8.2 What Stitchwort could take

1. Generated `f.cases` and `f.induct` on `derive_induct.rs`'s pattern (T1, D1910); `T.induct` itself is done.
2. ~~Derived `T.all` for enumerations~~: superseded by `decide.md`'s D1972.
3. `E-TERMINATION` suggests a `decreasing` clause (T3).
4. Automatic `attach` for calls inside list callbacks (T2).
5. Constructor short form resolved by expected type (T6, SW-510).
6. Audit runtime representations of `Fin`, `Vec`, `Member`, `Mod` (T5).
7. Record that sized types stay closed (SW-199 option D).
8. Motive inference by pattern-matching unification as a fallback (research).
9. Derived decision procedures and generators for inductive relations (research).
10. Partial correctness for `Partial` functions; coinductive propositions (research, not now).

### 8.3 Acted on, and open

The audit withdrew D1912: it rested on day 21 needing enumerations only, but its `DirectionKey` has a field. `T.induct` (D1823) and `auto`'s proof of `decreasing T.size p` from an attached element (D1450's ask) were built in nested types stage 3. Open: D1910 and D1911.

## 9 Infrastructure and trust

**Register:** `docs/research/infrastructure.md`. 68 screened, 29 carded, 29 read, 0 NOT READ; audited. Eris, the one card first left unread, was read from the PDF Matt fetched.

### 9.1 Key findings

- **Oracles from the corpus.** Hephaestus found 156 typing bugs (137 confirmed, 85 fixed) in javac, kotlinc and groovyc; 74 of them (52 inference, 22 soundness) came from two mutations of programs that already compile, which its generator alone missed: delete type annotations (must still compile) and replace a type by an incompatible one (must not) [24]. Thalia builds programs from library signatures and found 84 bugs, at least 42 of them missed by Hephaestus [122]. Stitchwort's elaborator fuzzer has neither oracle. Content hashes make the erasure oracle stronger here: removing an inferable annotation must leave the hash unchanged.
- **Error enumeration.** Eris replaces one expression of a well-typed seed with a term of an incompatible type, at every location, one mismatch per variant, for one representative of each *type shape*. Over ten months it found 64 bugs in kotlinc, scalac, groovyc and the Checker Framework's Nullness checker: 51 soundness bugs and 13 crashes, 33 of them fixed. Of the 51, 29 need a particular actual/expected pair at a particular location, and its authors judged 50 of the 64 out of Hephaestus's and Thalia's reach. Inference can "repair" an injected mismatch, so Eris recovers missing types from use sites (needed at 26% of locations, and for 27 of the 64 bugs); shapes cut the variants 13-fold [123]. Stitchwort's kernel sees only complete terms (`crates/core/src/term.rs:218-220`), so there the repair problem doesn't arise; the elaborator already records the expected type at each checked span for hover (`crates/elab/src/check.rs:3091`).

- **Generators.** Generate at the IR the backends share, as Rustlantis does with MIR (22 bugs) [136]; choose types non-locally so functions use their arguments (four times faster to find GHC bugs) [39]; score generators by mutants [120]; a second implementation is an oracle [119]; an LLM can seed inputs for an unfamiliar language [139].
- **Keep the reference evaluator in every comparison.** Dafny's compiler fuzzers found 24 new bugs, 9 of them soundness bugs; bugs in a shared front end make "all back-ends . . . exhibit the same incorrect result", so they added an independent interpreter [32]. Stitchwort's runtime and JS backend share their erasure decisions (`crates/rt/src/erasure.rs`), while `run --reference` uses the kernel's own erasure.
- **Fuzz the trusted base, not the proofs.** 105,823,818 executions against a zlib proved in Lean found no bug in the verified code, one heap overflow in Lean's C++ runtime and one crash in an unproved parser [58].
- **Cache keys.** Lean keeps proofs private to their module, so proof edits don't rebuild importers [82]; build-system vocabulary [101]; durability levels [115]; coarse roots invalidate everything [83]; incremental elaboration within declarations [75]. Stitchwort's tests run erased (`crates/driver/src/driver.rs:1241-1243`), so a test result keyed on the erased program can't be changed by a proof edit.
- **Agent-driven development.** "The task verifier is nearly perfect, otherwise Claude will solve the wrong problem" [21]. Agents under pressure edit tests; a "flag for human intervention" exit cut GPT-5's cheating from 54% to 9% [144]. Detailed type errors help agents fix code [63], and compiler feedback took GPT-5 from 22 to 54 of 56 Idris exercises [88]. Mathlib's bot lists renamed lemmas and debt metrics on each pull request [10]; Unison's renames are "same hash, new name" [134].
- **Editors and machines.** Type definitions are the most useful context for a model filling a hole [14]; LSP is the wrong interface for machines [9]; presentations that keep a reference to each subterm [104]; check what is on screen first [29]; total error recovery [143].
- Also: MutDafny mutates programs to find weak specifications (on average one in every 241 lines) [8]; Sisyphus repairs proofs, cheap only in its first, differential step [43]; verified extraction is safe only for first-order interop, which supports Stitchwort's first-order JS exports [38].

## 9.2 What Stitchwort could take

1. A lane trust diff: changed statements, renames, must-reject, snapshot and budget changes, axioms (T1, D1930).
2. An erase oracle, and error enumeration at every check site, kernel first, then the elaborator (T2, D2031; SW-686, SW-687).
3. A kernel type-preservation sweep: evaluate, read back, re-check (T3).
4. Rotate the sampled sweeps' offset (T4).
5. Score fuzzers against missed `cargo mutants` mutants (T5, D1932).
6. Weak-theorem detection by mutating functions (T6).
7. A detailed error mode for agent lanes (T7).
8. Goals list the data types they mention (T8).
9. SW-688 as an IR-level backend generator, always compared with `run --reference` (T9).
10. Test results keyed on the erased program (T10, D2030).
11. Measure cross-namespace proof unfolding before deciding opaque theorems (T11, D1931).
12. Deprecated aliases for std renames (T12).

### 9.3 Acted on, and open

The audit corrected Hephaestus's numbers (the 29 October card had the anonymous submission copy's) and found that tests run erased, which simplified T10 and produced D2030. Matt then fetched Eris; reading it turned T2's overwrite half into error enumeration and produced D2031. Open: D1930, D1931, D1932, D2030, D2031.

## 10 Compilation and runtime

**Register:** `docs/research/compilation.md`. 52 screened, 23 carded, 22 read, 1 NOT READ; audited.

### 10.1 Key findings

- **Most standard ideas are already in.** The October runtime survey's ideas (erased IR, register bytecode, reuse tokens, small numbers inline, TRMC, fused state handlers, Cranelift) are built.
- **Arrays with in-place update.** Lean's `Array.write` updates destructively when unshared [132]; a record of an efficient encoding plus an erased invariant and a view is the general method [7]; FP$^2$'s dynamic uniqueness test costs about 1% when nothing is reused [93]. The risk: in Huisinga's own `Array.groupBy` example, the array also sits in a map, so every `push` copies it ("accidentally quadratic"); the thesis presents this as its own illustration, not a bug in Lean's library [49].
- **Borrowing has costs.** Inferring borrows removes 75–100% of increments on 11 of 13 benchmarks against Perceus without reuse, 1.48× geomean, but blocked 6.4% of in-place updates in one benchmark [17]. Koka declined Lean's borrow inference as "not safe for space" [92]. In Lean's own ablation, removing borrowing made `const_fold` faster (0.90) [132]; Lean now lets users annotate borrows, with tail calls taking priority [75, 80].
- **Forced indices.** A constructor argument that appears in its result indices needn't be stored, saving 10–20% of memory on vectors [15]; Agda's GHC backend drops erased fields [6]. Stitchwort's `VCons` stores its length, and `vec.md` measured 5–10% per operation.
- **Dispatch is not the lever.** Tail-call dispatch in Rust ran 1.5–2× faster natively but 1.2–4.6× slower in WebAssembly engines, and needs nightly [54]; CPython's version gained 1–5% against a good baseline [33, 53]; Wizard's threaded dispatch, 14% on average [131]; Deegen has no ablation [140]; register bytecode was wasmi's big step [137]. Dispatch is about 19% of Stitchwort's machine time (`native-compilation.md` §1.2), so SW-813 should start from the call path.
- **Small numbers in JS.** The JS backend makes every `Nat` a `bigint`; day 2's brute force took 66.8 s under `--js` against 23.1 s on the machine. Scala.js measured `BigInt` longs 60 times slower than its own, in 2018 [129].
- **Measure with instruction counts** on a loaded machine [127].
- **Corner cases in shifts.** Verifying CertiCoq-Wasm found that WasmCert modelled Wasm shifts like C shifts, and a corner case that led to an unsoundness in Rocq's `vm_compute` [98].
- Not transferable now: lazy reference counting, which bounds pauses rather than saving time [65]; a Wasm backend, which still needs its own stack for deep recursion [138]. Context: lambda set specialisation is the general form of `opt.rs`'s known-argument copying [16]; zipper-based tree updates [94].

### 10.2 What Stitchwort could take

1. Benchmarks record retired instructions and bytecode counts beside time, and gate on instructions (T1; SW-659).
2. Shift and bitwise natives tested at and past the word width in every runner (T7; SW-808).
3. Count reference-count operations per run, the gate for any borrowing (T2, D1921).
4. `Array` specified as a list, in place when unique, with a copy counter (T6, D1920; SW-112).
5. SW-813: start from the call path; no tail-call dispatch while the playground is Wasm (T4).
6. Check reuse-token placement against Koka's fragility examples (T8).
7. JS `Nat` as `number` below $2^{53}$, `bigint` above, measured first (T5, D1922).
8. A "runs in place" report for std functions.
9. `Map` insert as a zipper loop that reuses every node (SW-814).
10. Borrowed parameters, only if item 3 shows a large share.
11. Don't store forced indices; drop self-passed relevant arguments.
12. A size counter for lowered code.
13. Unboxed scalar fields in cells, scalars after pointers.

### 10.3 Acted on, and open

The audit corrected C8 and two file references; nothing has been built yet. Open: D1920, D1921, D1922.

## 11 Cross-cutting themes

Each theme below appears in at least two registers; none is new to this summary.

### 11.1 Soundness hardening, and where real bugs were found

The kernel register's 2026 table (§5.4) was the most productive single exercise. Three of its shapes are closed by construction in Stitchwort and two by design, which the 14 tests confirmed; one (natives bound by name) is a known small gap. The bit-level shortcut had not been audited, and porting it, Lean4Lean's overflow shape, found SW-902. That matches Lean4Lean's observation that bugs live where nobody considered the error condition [22], and Stérin's that adversarial agents given past bugs find new ones [126]. The natives gap is SW-898. Two hazards were found before the features that would expose them: recursive propositions under proof irrelevance (D1980) and record eta over proof-only records (D1981). The infrastructure register adds the testing side: must-reject examples paired with accepted neighbours, a type-preservation sweep, and an "either" category for honest corner cases [74].

### 11.2 Proof automation a small, predictable kernel can afford

Every automation register finds the same shape: search outside the kernel, produce an ordinary proof term or certificate, and let the kernel check it. `grind` does this and reads only what it has collected [78]; Stitchwort's `omega`, `ring` and `cases` already do. External solvers buy a few points at the cost of large reconstruction libraries [48, 100], and Stitchwort's failures are about facts in scope, not large lemma libraries. Proposals that keep the kernel unchanged (congruence closure in `cases`, generated case theorems, write-back of found proofs) come first; the one proposal that would trust more, monad laws as conversion rules, is a design note only (D1850).

### 11.3 Elaboration order and postponement

Three registers meet here. Elaboration recommends one postponement queue [68, 106]; inductives and proof automation recommend motive inference from the expected type (`elab_as_elim`, pattern-matching unification [112]); SW-805's field-name rule is acceptable because it depends only on the text, not on inference order [46]. The common principle: anything that depends on inference order must wait in one place, with a fixed ladder of defaults, and anything decided from the text alone can run at once.

### 11.4 Testing oracles

Differential testing against an independent implementation is the oracle the literature trusts most: Rustlantis compares backends [136], fuzz-d adds an interpreter because shared code fools back-end comparisons [32], and Sisyphus runs old and new versions on the same inputs [43]. Stitchwort's reference evaluator is that implementation and must stay in every comparison. Mutation gives oracles without a reference: Hephaestus's erase and overwrite [24], Eris's enumeration of one mismatch at every location [123], MutDafny's weak-specification check [8], ETNA's scoring of generators [120]. The counterexample search is itself an oracle with a blind spot (SW-903), and the remedy is the same as for any oracle: report what it did not reach.

### 11.5 Trusted base

The registers keep returning to what is trusted: the kernel's re-check is the soundness argument for the lossy unifier; natives are trusted and should be pinned to the definitions they replace; R3/R4 are trusted conversion rules with a published confluence argument; the runtime, JS codecs and natives are where fuzzing finds bugs [58]; JS exports stay first-order because callbacks can be impure [38]; an erased test key would add trusted cache code, guarded by a differential test (D2030).

### 11.6 Measure before designing

Many recommendations are measurements: count reference-count operations before borrowing (D1921), time JS `number` against `bigint` (D1922), measure how often hypotheses hold only for small inputs (SW-903), survey existing proofs before congruence closure (D2000), count cross-namespace proof unfoldings before opaque theorems (D1931), and the postponement note's counter of queueable sites (1,073) before its design.

## 12 Recommendations

Table 3 merges the seven registers' rankings into one list. Priority weighs value (soundness first, then open tickets and friction in real programs) against effort, as the registers do; within a priority band the order follows the registers. "Open" means proposed and not started; done items are kept for the record.

Table 3: One prioritised list across all clusters.

| # | Item | Value | Effort | Ticket / decision | Status |
|---|---|---|---|---|---|
| 1 | Must-reject tests for 2026 kernel-bug shapes (kernel T1) | soundness | small | D1983 | done |

| # | Item | Value | Effort | Ticket / decision | Status |
|---|---|---|---|---|---|
| 2 | Fix universe-level overflow | soundness | small | SW-902, D1984 | done |
| 3 | §9.4 on the triangle property (kernel T4) | soundness argument | doc | D1983 | done |
| 4 | Refuse large elimination of recursive propositions (kernel T2) | soundness before SW-93 | small kernel | D1980; SW-93 | decided, with SW-93 |
| 5 | Record eta only with a non-proof field (kernel T3) | soundness after SW-93 | doc | D1981; SW-55, SW-789 | decided (re-minders) |
| 6 | Pin natives by hash or check their equations (kernel T5) | trusted base | small kernel | SW-898 | open |
| 7 | Lane trust diff (infra T1) | review of agent work | small tooling | D1930 | open |
| 8 | Counterexample search reports the size reached (PA T2) | silent misses | small | SW-903 | open |
| 9 | Erase oracle; error enumeration, kernel first (infra T2) | kernel and elaborator soundness | small (kernel), tooling | SW-686, SW-687; D1932, D2031 | open |
| 10 | Kernel type-preservation sweep (infra T3) | kernel soundness | small | — | open |
| 11 | One postponement queue (elab T1) | SW-817, 847, 885, 509 | elaborator | D1990, D2012, D2013, D2014; SW-817 | in progress |
| 12 | SW-805 field-name rule as name resolution (elab T3) | SW-805, SW-817 | small | D1991; SW-805 | done (rule) |
| 13 | Benchmarks gated on instruction counts (comp T1) | every speed lane | small | SW-659 | open |
| 14 | Shift natives at and past the word width (comp T7) | correctness | small tests | SW-808 | open (with SW-808) |
| 15 | Document and test the lossy rule (elab T2) | predictability | small | — | open |
| 16 | Score fuzzers with missed mutants (infra T5) | choosing SW-688/689 work | small | D1932 | open |
| 17 | Congruence-closure closer in `cases` (PA T1) | SW-608 | elaborator | D2000; SW-904 | evidence first |
| 18 | Generated `f.cases` and `f.induct` (PA T4, inductives T1) | SW-802, SW-809, SW-199 | elaborator | D2001, D1910 | open |
| 19 | Count reference-count operations (comp T2) | gate for borrowing | small | D1921 | open |

| # | Item | Value | Effort | Ticket / decision | Status |
|---|---|---|---|---|---|
| 20 | Test cache keyed on the erased program (infra T10) | warm tests after proof edits | small | D2030 | open |
| 21 | Rotating sample offset (infra T4) | coverage | small | — | open |
| 22 | Handler fast-path report (effects T5), after re-measuring SW-861 | SW-861, SW-791 | small | — | open |
| 23 | `E-TERMINATION` suggests a `decreasing` clause (inductives T3) | messages | small | — | open |
| 24 | Lookahead before `E-CASES-BUDGET` (PA T6) | SW-608 | small | — | open |
| 25 | Show what automation found; try automation at each `?` (PA T5) | readability, agents | small | — | open |
| 26 | smalltt probes; Courant–Leroy probes (elab T4, kernel T7) | proof-check speed | small | — | open |
| 27 | `decide`: extend `cases`, derive `Finite` | SW-800, SW-807 | elaborator | D1971, D1972, D1973 | awaiting review |
| 28 | Handler specifications as predicate transformers (effects T4) | SW-570 | design note | D1851 | open |
| 29 | `Array` specified as a list, in place when unique (comp T6) | SW-112 | design, runtime | D1920 | open |
| 30 | JS `Nat` as `number`, measured first (comp T5) | `run --js` | measure | D1922 | open |
| 31 | Automatic `attach` (inductives T2) | SW-199 | elaborator | — | open |
| 32 | Agent detail mode; goals list their data types (infra T7, T8) | agents | small | — | open |
| 33 | Monad laws as trusted conversion rules (effects T3) | SW-548 | kernel | D1850 | design note only |
| 34 | Cross-namespace proof unfolding, then opaque theorems (infra T11) | early cutoff | measure | D1931 | not yet |
| 35 | Codata, sized types, observational equality, universe polymorphism | — | research | D1911 | not recommended |
| 36 | SMT or ATP back ends; e-graph rewriting under binders | — | research | — | not recommended |
| 37 | Tail-call dispatch while the playground is Wasm; a Wasm backend | — | runtime | — | not recommended |

## 13 Open decisions for Matt

Table 4 lists every decision still open in the registers and design notes, with each register's recommendation and its assumptions in full. Each was checked against the board and `git log` on 30 October; resolved ones are left out (D1852, D1912, D1980, D1981, D1983, D1984, D1990, D1991, D2000 and D2012; the postponement note's D2010 was superseded by D2012, and its D2011 is kept and not listed there as open). D2013, D2014 and D2016 come from draft 3 of `docs/proposals/postponement.md`, which works in its external review; the reviewer agrees with D2013 and D2014. D1971–D1973 come from `docs/proposals/decide.md`, whose review page has been published and awaits an external review.

Table 4: Open decisions.

| ID | Question | Recommended | Assumptions, in full |
|---|---|---|---|
| D1850 | Should the monad laws of `Effect.bind` and `Effect.foldComp` hold by conversion (trusted kernel rules, as `Nat.add`'s two extra rules), though their equations need funext? | Yes to a design note (T3) only; decide the kernel change after it. | The decision line states none. The deep read (§4.1) names one: rules whose equations need funext are acceptable as trusted conversion (some theorems would stop listing `Quot.sound` in `stitchwort axioms`; consistency should be safe, because funext holds in the models Stitchwort's theory already assumes [INFERRED]). Its risks: a normalisation argument for the whole system must be written out; the elaborator's unification must apply the same rules. |
| D1851 | A specification layer for handlers in std (`wp` per handler, after Lean's `Std.Do`), aimed at SW-570 and rogue's loop? | Yes, design note first (T4), after T2 lands. | The decision line states none. The deep read (§4.3) names one: the cost is in symbolic evaluation of handlers, not in the property (rogue's notes support it). Its risks: a `wp` that recurses through a thunk will need a data form, as `AlwaysTree` did; partial programs still need fuel. |
| D1910 | Write one design note covering `f.cases` and `f.induct`, replacing D231–D233's "not now" and answering D1439? | Yes. | (A1) the elaborator keeps, or can recompute, the pattern variables, the index equations learned and the `if` conditions at each leaf of a compiled case tree; (A3) generated theorems are made only when a program names them. |
| D1911 | Keep codata out of Stitchwort for now (no streams, no coinductive types or propositions), and revisit only when a program needs infinite data? | Yes. | (P1) no planned program needs it; (P2) the `Partial` model stays a one-element set. Hyvernat's result means a later design needs its own totality checker, not a second size-change pass. |
| D1920 | For SW-112, write the design note around a list-specified `Array` with in-place update on unique values in the machine and native code, rather than a persistent tree or a uniqueness type? | Yes: design note T6; build only after it settles the JS choice. | Users thread arrays linearly most of the time; a copy counter makes the other case visible; JS may be slower. |

| ID | Question | Recommended | Assumptions, in full |
|---|---|---|---|
| D1921 | Spend a small lane counting reference-count operations (T2), and design borrowed parameters only if the count shows a large share? | Yes. | The count is cheap to add next to `alloc_trace`. |
| D1922 | Measure small `Nat`s as JS `number`s (T5) now, or leave `--js` as it is because native code is the CLI's speed route? | Measure now: JS is the faster runner on parse-heavy programs (SW-593). | The 2018 `BigInt` gap hasn't closed; T5 tests exactly that. |
| D1930 | Make the trust diff (T1) a standard part of reviewing every lane? | Yes: build T1, then add "read the trust diff first" to `docs/workflow.md`. | Agents keep writing most code and proofs; reviewing, not writing, is the bottleneck; the list is usually empty. |
| D1931 | Should theorems be opaque to other namespaces, so a proof edit stops at its namespace (Lean's module-system rule)? | Not yet: run T11 and T10 first, decide on the numbers. | T11 finds few cross-namespace proof unfoldings; std lemma edits are common enough to matter; an explicit opt-out like Lean's `@[expose]` is acceptable. |
| D1932 | For SW-686/687/688, do the corpus oracles (T2, T3) and the mutant scoreboard (T5) before building any generator? | Yes. | Mutating Stitchwort's large example and AoC corpus reaches more code per hour than a new generator, as the Rust results suggest. |
| D2030 | Key test results on the erased program (T10), so editing a proof never re-runs a test? | Yes, with the differential test; first measure how many tests a typical std proof edit re-runs. | (1) every test runs erased, on the fast runtime's erased IR or on the evaluator with erasure switched on (`crates/driver/src/driver.rs:1229-1243`; tests of partial code too, `crates/driver/src/run.rs:545`), so no proof body or type argument is computed and a proof edit can't change a test's outcome; (2) the erased hash of a test can be computed from `erase.rs`'s decisions, with each reference hashed by that declaration's own erased hash, and it changes whenever anything that runs changes, including a definition's body, a data type's constructors and the evaluation modes already in the key; (3) a second hash per declaration in trusted cache code is acceptable if a differential test (the same results with either key, over the test corpus) guards it, as `check-cache.md` §5.4 guards the check cache; (4) the saving matters, because std lemma edits re-run many tests today (unmeasured). |

| ID | Question | Recommended | Assumptions, in full |
|---|---|---|---|
| D2031 | Build T2's overwrite half as error enumeration (Eris): every check site, one replacement per variant, one term per shape of incompatible type, kernel stage first, run as an on-demand campaign with a small fixed-seed daily sample, instead of random overwrites in the daily `fuzz_elab` run? | Yes. Stage K (the kernel) first: small, needs no type recovery, tests the trust boundary; the elaborator stage after T2's erase half, which shares its harness. | (1) Eris's evidence holds in kind: soundness bugs in mature checkers need a particular actual/expected pair at a particular location (29 of its 51), so random overwrites reach few; it comes from four JVM checkers with subtyping, inference and flow typing, and its transfer to a dependently typed checker with conversion and no subtyping is untested; (2) no metavariable reaches the kernel (`crates/core/src/term.rs:218-220`), so a replacement whose type the kernel's own conversion check finds not convertible to the expected type makes the definition ill-typed and must be refused; a comparison out of budget drops the variant; (3) the elaborator's hover record (`crates/elab/src/check.rs:3091`, `crates/elab/src/nav.rs:503-519`) can be extended to mark check-mode spans and whether their expected type still had an unsolved metavariable, and the first elaborator stage skips those, so inference can't turn a mismatch into a false alarm (Eris's conservative baseline lost 26% of locations and 27 of 64 bugs, which use-site recovery would win back later); (4) everything the elaborator accepts beyond unification (ability coercions and total-to-partial lifting, postponed row comparisons, numeral defaults: `compare`, `crates/elab/src/check.rs:4010-4084`) counts as compatible when choosing incompatible types; (5) the cost is acceptable outside the tiers (thousands of variants per seed, minutes per seed at about 50 ms a check), and the daily sample guards against regressions; (6) every acceptance is checked by hand before it is called a bug, and each confirmed one becomes an `examples/soundness/` file with its expected code; (7) the tiers' time budgets change only by the daily sample. |
| D2001 | For SW-802 and SW-809, design generated case theorems (`f.cases`, Lean's `fun_cases`) instead of per-leaf equation lemmas (D231)? | Yes. | (F1) the elaborated case tree records which equation each leaf tested, readably enough to state the branches; (F2) today's `cases` does not already prove the day 16 and day 23 statements that failed with `E-MATCH-MOTIVE`; T4 checks this first, and if `cases` does, the decision lapses to "document `cases` for this". |

| ID | Question | Recommended | Assumptions, in full |
|---|---|---|---|
| D2013 | Run SW-805's field-name rule as a default in the queue's ladder, and default number literals in the same ladder after it, instead of at the literal? | Yes. | (R1, from D1991) the field-name rule looks only at the literal's field names and the names in scope at the literal, never at types learned later; running it later doesn't change what it looks at, only whether something else decided first. (N1) a number literal that waits ends with the type the rest of the declaration fixes, and `Nat` where nothing does; so in a program accepted today, where each such literal was made `Nat` and everything else agreed with that, it still ends as `Nat`; the change only accepts programs that fail today because a literal was made `Nat` too early; stage 4 tests this over the corpus (inferred, not yet measured: K4 is not in the prototype). (N2) the field-name rule has priority over the number default because a record's field types say what its numbers are, never the reverse (§5.4's "in a list"). (N3) Lean's numeral default is the precedent: low priority, in the ladder (`Prelude.lean:1411`, `SyntheticMVars.lean:237-243`). The external reviewer agrees: record identity first, `Nat` last. |
| D2014 | Resume queued checks at the end of every application, scoped to the items made inside it, as well as before lifting and at the end of the declaration? | Yes. | (S1) Stitchwort's unifier can't wait: a comparison that meets an unfilled placeholder answers `false` at once, and its caller takes another route, so a placeholder must be filled before the comparisons that would need it; the end of the application that created it is the first fixed point after the later arguments that may solve its blocker, and before the application's type is compared with what is expected of it; the prototype shows the end of the declaration is too late for std (`Map/Spec.sw:1862`), and the end of each application is enough for the whole corpus. (S2) scoping to the application's own items keeps a failing item from being blamed on, and undone with, an attempt it doesn't belong to, as Lean's `withSynthesize` does; items from outside are resumed at their own application's end or at the declaration's. (S3) waking inside the unifier would elaborate syntax in the middle of a comparison that callers inside many of the 44 `Snapshot::take` sites branch on; resume points give the same order at fixed places. (S4) if waiting comparisons (§5.9) are ever added, W1 (the resume point at each application's end) can become less frequent; this decision doesn't preclude that. The external reviewer agrees, and calls scoping non-negotiable: unscoped, the prototype accepted a program that today's order rejects. |

| ID | Question | Recommended | Assumptions, in full |
|---|---|---|---|
| D2016 | Keep placeholders solvable by unification until stage 6 (comparisons that can wait), and freeze them only then? | Yes (draft 3, against the reviewer's suggestion to freeze them now). | (P1) a solution unification finds for a placeholder is forced by the comparison that found it, not a guess; (P2) the fill compares the item's own term with any such solution, so a disagreement is reported as the item's own error at its span; (P3) zonking writes the item's term, never the unifier's solution; (P4) a frozen placeholder needs comparisons that can wait, and without them they fail on correct programs: freezing broke `lambda-postponed-dependent.sw` and four AoC proofs (2024 days 9 and 24), while in 14 of 135 fills unification had already solved the placeholder. |
| D1971 | Close SW-800 by extending `cases` (a decision closer at the leaves, and splitting a finite variable when nothing is stuck), with no new `decide` word? | Yes. | `decide.md` gives the alternative rather than assumptions: a `decide` word evaluating `Decidable` instances, as Lean does, can't do day 21's `tabulateSame` and doesn't show the failing case. The proof-automation register finds that the literature supports it. |
| D1972 | Build SW-807 as a derived `Finite` interface (a list of every value with a completeness proof) and `Finite.check`, for non-recursive constructor types with finite fields? | Yes, after D1971's conversion of days 12 and 21. | Alternative given: enumerations only, as Haskell's `Enum`; day 21's `DirectionKey` has a field and would be left out (confirmed by the inductives audit). |
| D1973 | Never add native evaluation as a proof (`native_decide`)? | Never, until a proof needs more than about 10 s of kernel evaluation; then revisit with that proof. | Alternative given: trust the fast runtime for closed `Bool`s, adding it to the trusted base. |

D1850 and D1851 state no assumptions in their decision lines; the table gives the assumptions and risks their deep reads (effects register §4.1 and §4.3) name.

# 14 Limits

## 14.1 Not read

Six carded sources were not read. None supports anything in this summary.

- Proof automation C26: Morrison and de Moura, *grind: an SMT-inspired tactic for Lean 4*, IJCAR 2026 (no public copy found; its system was read through its manual, source and slides).
- Proof automation C27: Rossel et al., *Towards Pen-and-Paper-Style Equational Reasoning in Interactive Theorem Provers by Equality Saturation*, POPL 2026, doi 10.1145/3776667 (ACM 403; its tool's README now points to `grind`).

- Proof automation C28: Bhat, Böving et al., *Interactive Bitvector Reasoning using Verified Bit-Blasting*, OOPSLA 2025, doi 10.1145/3763167 (no bit vectors yet).
- Proof automation C29: Limperg and From, *Aesop*, CPP 2023, doi 10.1145/3573105.3575671 (no tactic language to search over).
- Kernel C27: Liesnikov and Cockx, *Building a Correct-by-Construction Type Checker for a Dependently Typed Core Language* (Agda Core), APLAS 2024, doi 10.1007/978-981-97-8943-6_4 (TU Delft repository unreachable).
- Compilation C22: IOG, *Stacks in the JS backend* (2023), a blog post that returns 404.

### 14.2 For Matt to fetch

Nothing is left. The one paper that met the bar, Eris [123], was open access at ACM but refused every scripted route; Matt fetched it on 30 October and it was read in full (Section 9). The figures its abstract gave, which the first version of this summary marked unverified, match the paper.

### 14.3 What wasn't searched

Each register's §9 lists what it did not reach. In short:

- **Effects**: Unison's and Koka v3.2's release notes, and the F⋆ indexed-effects venue, were not checked; the rows they back stay unverified. The 64 screened-only rows were not re-screened.
- **Elaboration**: in-window work on elaborating effect rows, incremental elaboration, Agda's and Idris 2's elaborator changelogs, RocqPL and Lean workshop programmes.
- **Proof automation**: ITP, CPP, IJCAR, CADE and FroCoS programmes year by year; Isabelle's Quickcheck and Nitpick after 2022; Agda's and Idris 2's automation.
- **Kernel**: Idris 2's core and its bugs; Agda's kernel bug history; small trusted kernels as a topic; hash-consing in production kernels beyond Courant and Leroy.
- **Inductives**: Idris 2's totality changelog, Rocq 9 fixpoints, F⋆ and Liquid Haskell termination, views papers, non-empty lists, or-patterns; venue sweeps of TyDe, CPP, ITP, FSCD and TYPES. One query (Dafny's default `decreases`) was never run.
- **Infrastructure**: MetaRocq updates, Agda interface files, F⋆ tooling, Nix content-addressed derivations, Dhall's pinned imports, Elm's semantic versioning, and the CoqPL, Lean workshop and ITP/CPP programmes year by year.
- **Compilation**: the search budget ran out after 36 queries on 29 October, so Unison's and Idris 2's backends, MLKit, deforestation, Koka's recent benchmarks, persistent vectors, Larose et al.'s AST-versus-bytecode paper, Lorenzen's thesis and Roc's docs were not screened.

### 14.4 What is inferred, not run

The registers mark their reasoning [INFERRED]. The main inferred claims behind the recommendations are:

- that congruence closure would close probes 1 and 2, and that `J`-based congruence steps type-check for the dependent calls that matter (D2000's A1–A3);
- the critical pairs of the proposed monad-law rules, worked by hand, with one overlap (R-fuse against R-assoc) still to check properly;
- that `f ?a = f x` with `f (Succ n) = f n` picks `?a := x` ("inferred, not run");
- that a record of proofs becomes unit-like under D1′ and breaks transitivity;
- that Stitchwort's size-change checker would reject `f (C2 x) = f (C1 x)` ("inferred, not run");

- the expected gains of borrowing (0–10%, possibly negative), of forced indices and of JS `numbers`, all unmeasured;
- that a test key on the erased program changes whenever anything that runs changes (D2030 (2)).

What *was* run: the proof-automation probes (three `cases` failures and the counterexample miss), the 14 must-reject programs, the parallel-rule test, the universe-overflow reproduction and fix, and the postponement prototype.

### 14.5 This summary

It cites only works whose cards say "Full text: read". Its numbers are taken from the registers as they stand after the 30 October audits. It did not re-read any paper. Where a register's file:line reference had moved on `main`, the current line is given here (for example `apply_args_postponing` is now at `check.rs:1903` and `CallMemo` at `value.rs:2334`). Bibliographic fields missing from a card were looked up in the arXiv API, Crossref and DataCite; no DOI was guessed, and entries that couldn't be completed carry a note.

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